CHAPTER 2

Comparative study of gallstone with urinary stone and kidney stone

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Introduction

Gallbladder disease is the most prominent yet neglected disease of the 21st century. Moreover, the disease has its own distinct complications, which further makes the treatment of these stones arduous. The prime impediment of this disease, which poses a challenge when tackling it, is the lack of symptoms (Behari & Kapoor, 2012). The key to permanently curing any disorder of the human body is the correct diagnosis of the disease. Failure in the diagnosis of the disease leaves the option of only managing the problem and not curing it, depending on the degree of severity of the disease. The patient suffering from gallstones or any other urinary stones does not experience any symptoms until the stones reach a reasonable size. Upon reaching a reasonable size, the patient experiences intensifying pain and, only then it is possible to diagnose the disease. At this particular stage, when the patient is experiencing intensifying pain and the stone has reached a larger size, this simply implies that medical scientists are only left with the option of surgically removing it (Diehl, 1992). Moreover, the lack of health awareness and lack of regular body checkups further aid in worsening the condition (Krogsbøll et al., 2019). Over time, the rough surfaces of these stones start damaging the sensitive walls to their origin organs, which creates a crisis situation. The continual damage of these walls leads to the failure of organs and even death in extreme cases. Prognosis is the key to control any disease from reaching its extremities.

The occurrence of gallstones in the human body is rising at a rocketing pace (Stinton & Shaffer, 2012). Statistical data represented by the United States gives horrifying numbers of diagnosing about 0.5 million patients

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with gallstones (Stinton & Shaffer, 2012). The research studies conducted concluded that 80% of gallstones are left undiagnosed due to the delayed onset of symptoms in patients (Miyaoka & Monga, 2009; Sakorafas et al., 2007). These numbers show the grave gravity of the condition faced by the population, which is unaware of its occurrence in their bodies.

To find a potential treatment for treating gallstones, it becomes necessary to understand their roots. The following sections will highlight every aspect of urinary stones and unfold the real reason behind their formation in the human body. The surgical options only remove the gallstones or any other urinary stones from their original site, but they can reappear in the patient during their lifetime (Lammert et al., 2016). We will extend the discussion by acquainting ourselves with the chemical compositions and mechanisms of the pathogenesis of these stones and will try to compare these findings.

Urinary stones

Lithiasis in the urinary tract is a crucial and painful condition that is widely affected among the population around the globe (Afdhal, 2004; Bernhoft et al., 1984; Nagaraja Rao et al., 2011; Rains, 1964; Ramaswamy et al., 2015; Trotman et al., 1974). Numerous hypotheses have been presented explaining the mechanism involved in the pathogenesis of urinary stones; still, the exact information about their genesis is under discussion. In the last few decades, the fundamental objective of research scientists has been to elucidate a correlation between the molecular constitution of the formed stones and the abnormalities observed in the patients (Nagaraja Rao et al., 2011; Ramaswamy et al., 2015). The perspective adopted by the scientists is that they study the relative levels of trace elements in the human body, and as per their levels, they deduce any pathological changes in the body (Tapiero et al., 2003). It is proven that changes in levels of trace elements are indicative of the pathological condition of the body. The genesis of stones can even lead to fatalities if ignored. These stones, especially gallstones, are known to disrupt the digestive tract, and kidney stones are known to gravely injure the urinary tract and the kidneys (Alok et al., 2013).

Gallstones

Gallstones are primarily one of the most painful conditions observed in the biliary tracts. They are basically the accumulations of cholesterol and

bile pigments that are formed in the gallbladder of the human body. They are formed when the stored biliary fluid in the gallbladder hardens into a stone-like hard mass (Afdhal, 2004). The liver hepatocytes secrete a greenish or yellow bitter fluid that is stored in the gallbladder for the digestion of food in most vertebrates (Afdhal, 2004; ; Rains, 1964;). This gallbladder wall then undergoes contractions that push this stored biliary fluid further into the bile duct, which is then pushed forward into the small intestine, where it induces its fat digestion role on the food ingested by the organism. The constituents of the biliary fluids are primarily cholesterol, fats, bile acids, bilirubin, and water (Afdhal, 2004; Bernhoft et al., 1984; Ramaswamy et al., 2015; Trotman et al., 1974). In a normal human body, the biliary fluid can completely dissolve the cholesterol. If there is an excess of cholesterol in the gallbladder, the bile fails to dissolve it completely, which leads to the formation of crystals and eventually stones in the gallbladder. In some cases, high levels of the pigment bilirubin are also observed in the biliary fluid, contributing to the etiopathogenesis of the gallstones. There are different types of gallstones prevalent among the population, each with its own distinct mechanism of formation. The internal constitution of gallstones is affected by a variety of factors like weight loss, diet, obesity, geographical region, and ethnicity.

Types of gallstones

The gallstones can be classified based on their chemical composition:

1. Cholesterol Stones
2. Pigment Stones
3. Mixed Stones

Cholesterol gallstones

These are the most prevalent types of gallstones observed in the population. As the name suggests, they are primarily comprising cholesterol. The white crystalline powder, namely cholesterol, is secreted by the liver hepatocytes into the bile, which is a mechanism of the human body to eliminate excess cholesterol from the human body (Stinton & Shaffer, 2012). If these hepatocytes secrete a large amount of cholesterol as compared with the amount of bile acid and lecithin, the bile is unable to dissolve such a large amount of cholesterol, which results in some amount of cholesterol being undissolved in the gallbladder. Also, if the hepatocytes fail to secrete enough lecithin and bile acids, they contribute to the insolubility of cholesterol. In either of these cases, these undissolved cholesterol

particles stick together and eventually grow in size, which leads to the formation of cholesterol gallstones (Di Ciaula & Portincasa, 2018).

Pigment stones

These stones are primarily composed of calcium bilirubinate, carbonate, phosphate, and other anions. The pigment from which it is composed is basically a waste product obtained from hemoglobin. Hemoglobin is a protein that carries oxygen to all cells of the body through red blood cells (RBCs). The old hemoglobin from the RBCs is destroyed and changed into bilirubin, which is released into the blood. This released bilirubin in the bile is eliminated from the blood by the liver, which converts it into modified bilirubin in the bile. If an excess of bilirubin is present in the bile, it causes other components of the bile to produce brown pigments. These pigment molecules are insoluble in the biliary fluid and inevitably aggregate to form stones of black and brown color. The brown pigment stones formed are relatively softer than the black pigment stones.

Mixed stones

As the name indicates, these stones are a combination of cholesterol and pigment stones, comprising varying relative proportions. They are also composed of other components like calcium stearate, calcium palmitate, bilirubin, and calcium carbonate. They are often found to be radiographically visible owing to their elevated calcium contents (Murphy et al., 2020).

Pathophysiology of gallstones

The mechanism involved in the pathophysiology of gallstones is quite complicated due to the involvement of numerous factors controlling their formation. The event that triggers the formation of these stones is the biochemical compositional changes in the bile, which lead to cholesterol supersaturation, leading to the genesis of gallstones. These events then culminate in the formation of gallstones in the gallbladder. These compositional changes lead to the disruption of the balance between anti-nucleation and nucleation-leading factors in the biliary fluid, thereby eventually causing abnormalities in the functioning of the gallbladder. The prerequisite for the solute to be separated out for the formation of crystals is that the solute present in the solution must be in a state of supersaturation; this can be deduced from the knowledge of crystallogeny. These conditions also suggest that the system under observation must be unstable considering

the laws of thermodynamics. The formation of these crystals involves two important steps: nucleation and crystallization. It primarily forms a crystalline nucleus comprising hundreds of molecules, leading to the crystals growing in size. In a solution, the nucleation process often requires a very high degree of saturation; this state of the solution that is supersaturated is known as the unstably supersaturated solution, also known as homogenous nucleation. Upon the addition of specific nucleating agents, we can induce nucleation in a comparatively low supersaturated solution as well. This type of induced nucleation is known as heterogeneous nucleation, and the solution is known as a substably supersaturated solution. To achieve homogenous nucleation of the biliary cholesterol, an extremely high degree of supersaturation is expected, which is approximately 300%, and it is practically impossible for such a large saturation to exist in the human body. This proves that the dominant process for the formation of gallstones is heterogenous nucleation, in which the biliary fluids act as a substably supersaturated solution and the crystals of bile should be acting as nucleating factors.

Table 2.1 denotes the chemical compositions of the various types of gallstones observed among the human population.

Fig. 2.1 illustrates the mechanism of the etiopathogenesis of gallstones.

Kidney stones

Kidney stones, most commonly known as renal calculi, are crystalline aggregations found in the kidneys of the human body. Many times, kidney stones of comparatively small size leave the human body during micturition without the occurrence of symptoms in the whole process. If the size of the kidney stones exceeds a critical limit of 2–3 mm, they become

Table 2.1 Chemical compositions of the various types of gallstones observed among the human population.

Chemical constituents	Cholesterol gallstone (mg/%)	Mixed gallstone (mg/%)	Pigmented gallstone (mg/%)
Cholesterol	84 ± 5.7	62.0 ± 4.2	17.7 ± 3.9
Calcium	0.9 ± 0.2	5.6 ± 1.1	28.0 ± 2.9
Phosphate	0.2 ± 0.2	10.6 ± 1.0	18.4 ± 1.0
Bilirubin	3.0 ± 1.4	9.0 ± 0.2	18.5 ± 2.6
Triglyceride	7.2 ± 0.8	6.2 ± 0.3	6.4 ± 0.3
Bile Acid	4.5 ± 0.2	7.0 ± 2.3	10.4 ± 2.3

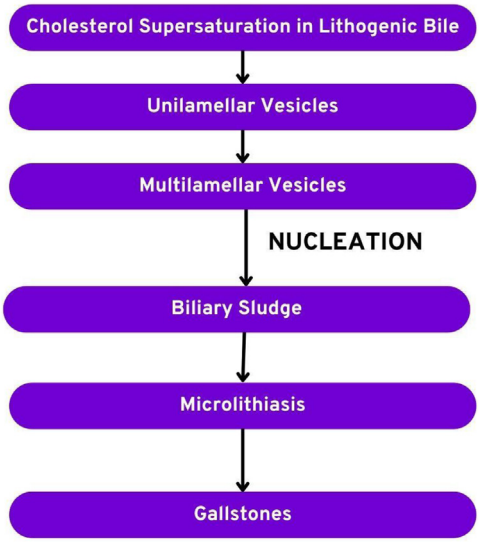

Figure 2.1 ***The mechanism of the etiopathogenesis of gallstones.*** This flow diagram explains the flow of the events that lead to the formation of gallstones.

an obstacle in the ureter, hence hindering their path to pass through the urinary system naturally. These are basically the accretions of the minerals dissolved in the urine. Their size is highly variable, ranging from millimeters to a few centimeters. The obstruction of the ureter by the kidney stones leads to dilation or stretching of the renal pelvis and ureter in addition to the spasms of the concerned muscles. During the passage of stones through the urinary system, episodic pain is observed that is usually felt in the flank of the abdomen on the lower side of the human body.

The occurrence of kidney stones is heavily influenced by genetic predisposition and various geographical factors associated with the patients. Another factor playing a major role in their occurrence is the dietary habits of the patients, along with the intrinsic metabolic rate, overall water intake, and hygiene of the human body. Severe abdominal pain is observed owing to the mechanical damage caused to the tissues of the organs through which it passes through the movement of kidney stones. The intensity of the abdominal pain can be directly correlated with the roughness of the surface of kidney stones and their relative size. According to the research study on the passage of kidney stones through the urinary tract, it was observed that the renal calculi having a size less than 10 mm pass through the passage instantaneously, whereas the passage

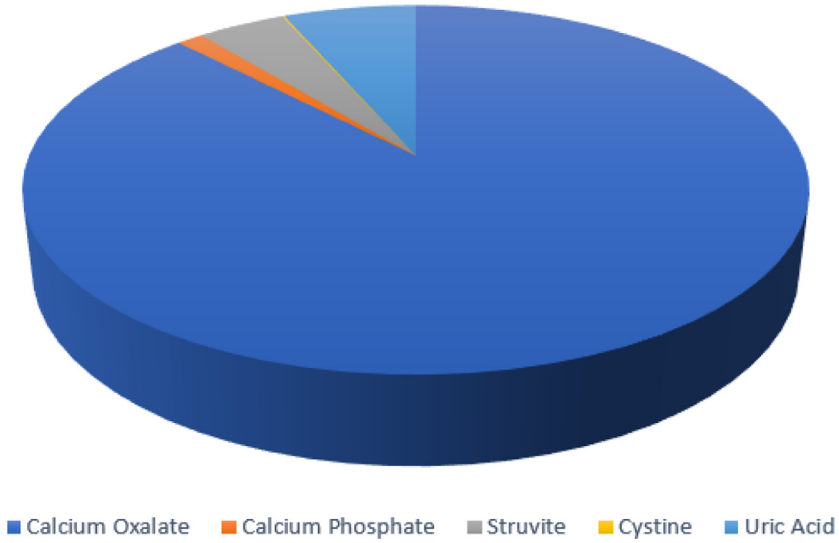

Figure 2.2 ***Constituents of kidney stones.*** This pie chart explains the overall chemical constitution of kidney stones.

of stones having a size greater than 10 mm often needs the aid of surgery to be completely eradicated from the human body; otherwise, it may block the ureters further, causing damage to the preexisting normal processes of the human body. The probability of the recurrence of renal calculi is 50%, and the preventive measures extended to the patient are solely decided based on the chemical disposition of the kidney stones.

Kidney stones exhibit a wide array of types based on their chemical makeup.

Fig. 2.2 illustrates the components found experimentally in kidney stones.

Table 2.2 disseminates a brief overview of the physiology of the renal calculi.

Calcium containing stones

The majority of kidney stones share this type of chemical composition, which involves calcium combined with phosphate, oxalate, or rarely uric acid. They occur in the forms of calcium carbonate, brushite, calcium oxalate, gypsum, dolomite, and calcium phosphate. Kidney stones with these kinds of compositions are found to be opaque to X-rays and similar kinds of radiation, thereby creating a hurdle in their diagnosis.

Table 2.2 Brief overview of the physiology of the renal calculi.

Etiology	Types	Symptoms
The predominant reasons behind the aggregation of renal calculi are Dent's disease, modularly sponge kidney, hyperparathyroidism, and renal tubular acidosis	Crystals of calcium oxalate comprise 80% of the total composition	Hematuria is the presence of blood in the urine due to damage to the internal walls of the kidneys and urethra.
	Crystals of urate (uric acid): 5%–10%	Colicky Pain
	Crystals of struvite: PO_4, Mg, and NH_4 crystals	Dysuria involves a burning sensation during micturition
	Crystals of cystine and calcium phosphate	Pyuria is the presence of pus in the urine
		Hydronephrosis and nausea
		Post-renal azotemia is the blocking of the ureter by kidney stones
		Oliguria which is the reduced volume of urine

Uric acid stones

These stones comprise magnesium ammonium phosphate, better known as struvite, and a crystallized solution of uric acid. Upon experimental analysis, these stones were found to be radiolucent. The possible reasons behind the formation of stones with this specific composition can be attributed to the precipitation of xanthine, uric acid, and adenine under low pH conditions.

Struvite stones

These stones are also known as triple phosphate stones. The primary reason behind their formation is infection, and hence they are known as

infection stones. The findings of the cross-sectional studies carried out on these stones suggested the presence of layers of white and brown color inside them. These are formed due to the presence of a hyperconcentration of trivalent phosphate and ammonium present in the urine. These stones are more prevalent in women than in men.

Cystine stones

These stones are formed due to the leakage of cystine in the urine. These stones create significant problems in patients because they have a tendency to get stuck in the walls of the kidneys or urinary tract. The physical appearance of pure L-cystine stones exhibits a yellow spherical structure with a homogenous appearance (Dal-Moro et al., 2005).

Protease-related stones

These stones are observed in HIV-positive patients. Their occurrence is involved with the administration of indinavir sulfate, a protease inhibitor employed in the treatment of HIV. These stones fail to be detected during spiral CT scanning.

Fig. 2.3 portrays the steps involved in kidney stone genesis.

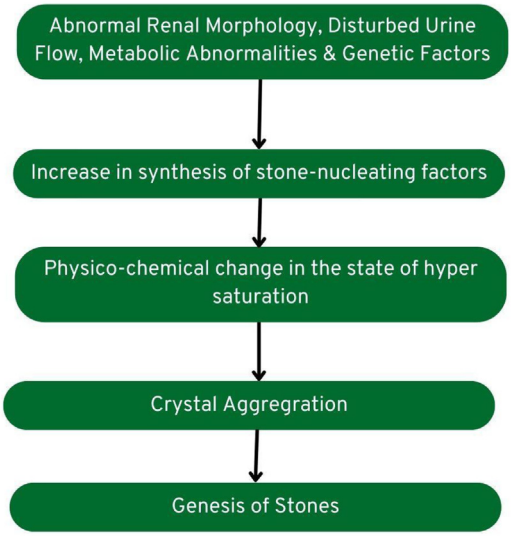

Figure 2.3 ***The mechanism of etiopathogenesis of kidney stones.*** This flow chart explains the flow of events that lead to the formation of kidney stones.

Comparative analysis of gallstone with kidney stones

There are appreciable differences observed among both the stones starting from their origin, chemical composition, causal factors, etc. These distinct differences help us understand the independent methods of treatment extended to the patients. Along with differences, one can also find similarities between them. The comparative analysis proposed in this chapter significantly helps to solve the numerous hurdles faced while treating these stones among the large affected population by deeply understanding the unclear mechanism of these disorders.

The comparative analysis is further categorized based on various contrasting characters as well as similar characters. This comparative analysis makes the understanding process more structural and precise, which eventually results in increased precision among the derived conclusions from these studies.

Gender prevalence

This is a neglected yet important aspect when considering comparative analysis. The physiology of men and women is different, which creates a need to extend a signature treatment that resonates with the body structure of the particular patient.

Fig. 2.4 illustrates the comparative analysis based on the gender occurrences observed among the human population.

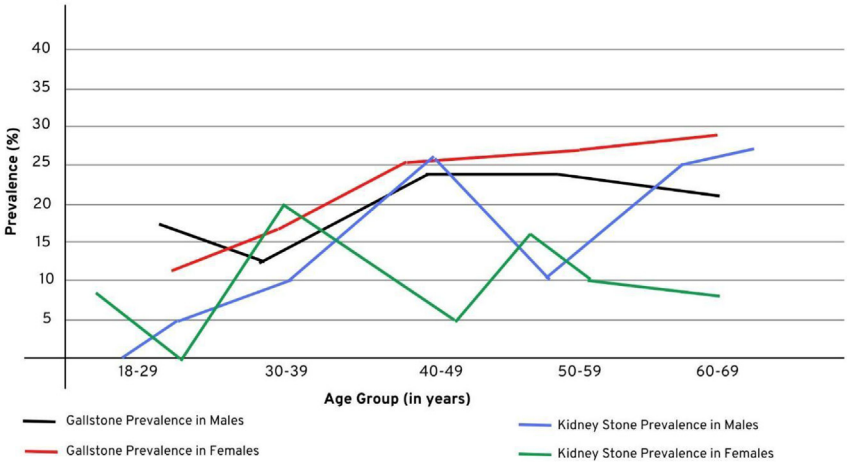

Figure 2.4 ***Comparative analysis based on the gender occurrences as observed among the human population.*** This line diagram analyzes the comparative gender prevalence among the human population.

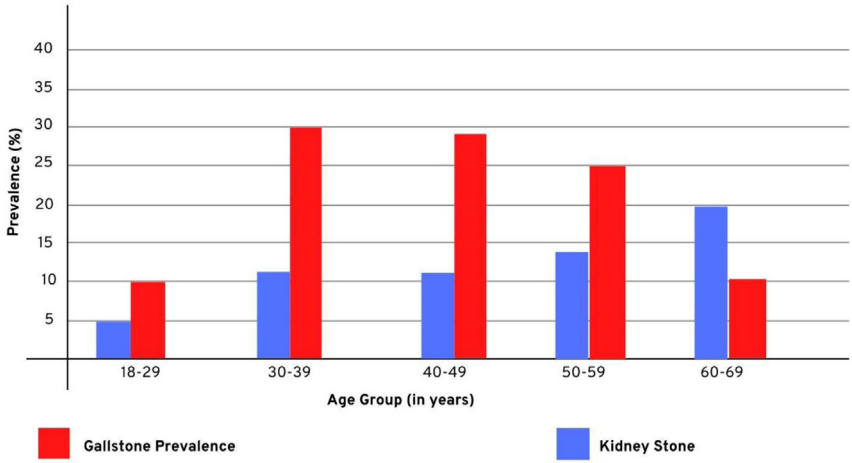

Figure 2.5 ***Comparative analysis based on the occurrence of gallstones and kidney stones in different age groups.*** This bar diagram analyzes the comparative prevalence of gallstones and kidney stones among varied age groups. It suggests a high prevalence of gallstones among the 18–29; 30–39; 40–49; and 50–59 age groups. A high prevalence of kidney stones is observed in the 50–59 age group.

Age prevalence

As the age of the human body progresses, its ability to fight infections and maintain its homeostasis becomes quite difficult as immunity is on the verge of decline, especially during old age. The thymus is in charge of immunity; thus, organ starts shrinking in size as the age of the body progresses, and thereby, its ability to provide immunity also regresses with its size.

Fig. 2.5 portrays the analysis based on the occurrence of gallstones and kidney stones in different age groups.

Prevalence based on types of stones

This aspect gives insights into the type of gallstone or kidney stone that is most predominant among the stones in the population.

Figs. 2.6 and 2.7 exhibit the relative distribution of gallstones and kidney stones respectively based on their types.

Chemical composition

The differences or similarities observed in the chemical composition of gallstones and kidney stones give us direct hints regarding the direction of treatment that must be administered to the patient based on their physiology.

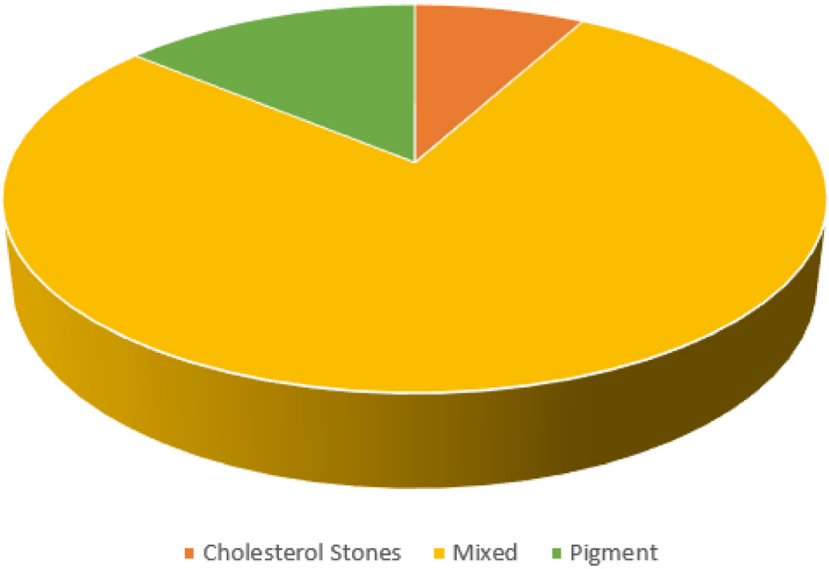

Figure 2.6 ***Relative distribution of types of gallstones observed in the human population.*** Cholesterol Stones occupies the highest percentage of occurrence among the human population followed by Pigment and Mixed Stones.

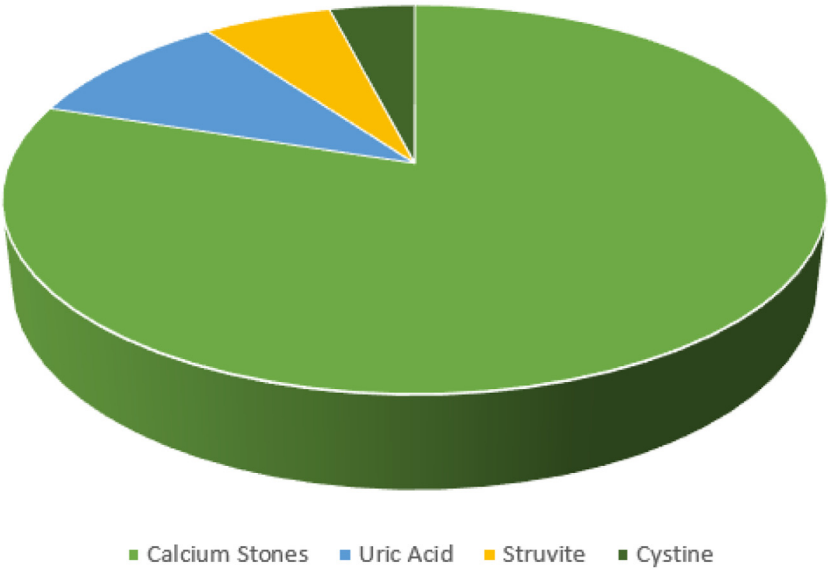

Figure 2.7 ***Relative distribution of types of kidney stones observed in the human population.*** Calcium Stones are the most prevalent among the varied types of kidney stones in the human population followed by Uric Acid Stones, Struvite Stones, and Cystine Stones.

Table 2.3 Similarities and differences observed in gallstones and kidney stones.

Parameters	Gallstones	Kidney stones
Origin	Gallbladder	Kidneys
Main constituent	Cholesterol	Calcium oxalates and salts
Promoting factor	A diet rich in fat and cholesterol	A diet rich in oxalates, sodium, and animal protein
Area of pain	Mid-upper abdomen	The lower back side of the abdomen and groin

The field of medical science is now evolving at a faster pace with ever-changing paradigms, which suggest the administration of personalized treatments should be given to other patients based on their genetic predisposition. Table 2.3 displays the differences and similarities observed in the chemical constitution of these stones, along with their proportions.

Table 2.3 overviews all the similarities and differences observed in gallstones and kidney stones.

Conclusions

The highly evolving medical field has instigated the need to ascertain treatments by analyzing the patient's genetic, physical, mental, and emotional states, which makes the whole treatment process personalized and highly effective. These changing paradigms force us to deeply understand the roots of the causes of various disorders that occur in the human body. Gallstone disease is one such condition in the medical sciences in which we don't have answers to many of the important questions that are directly associated with the treatment of this disease. The comparative analysis and the explanation of the pathogenesis of gallstones and kidney stones help us realize the symmetry in nature. These differences and similarities help us to clear the ambiguities that are persistent in our mind, which create obstacles in our path of ascertaining treatments for this disease. The characteristic of gallstones—the absence of symptoms—is a very fatal consequence of this disorder, which is often neglected, but by skimming through their prevalence rates, the gravity of this condition can be understood. This area requires more research, mainly to understand the failure of oral medications to act on the gallstone of appreciable size. The future of medical science will be highly focused on administering treatments that focus on the genetic makeup of the patient, thereby combating the roots of the disease.

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